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CHE 330

Design of Experiment[†]

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Design of experiment or experimental design is a structured approach that could be used to establish causality relationship between factors (along with their possible interaction effects) and the outcome of an experiment. The approach is general with broad applications in engineering, natural science, social science, and finance.

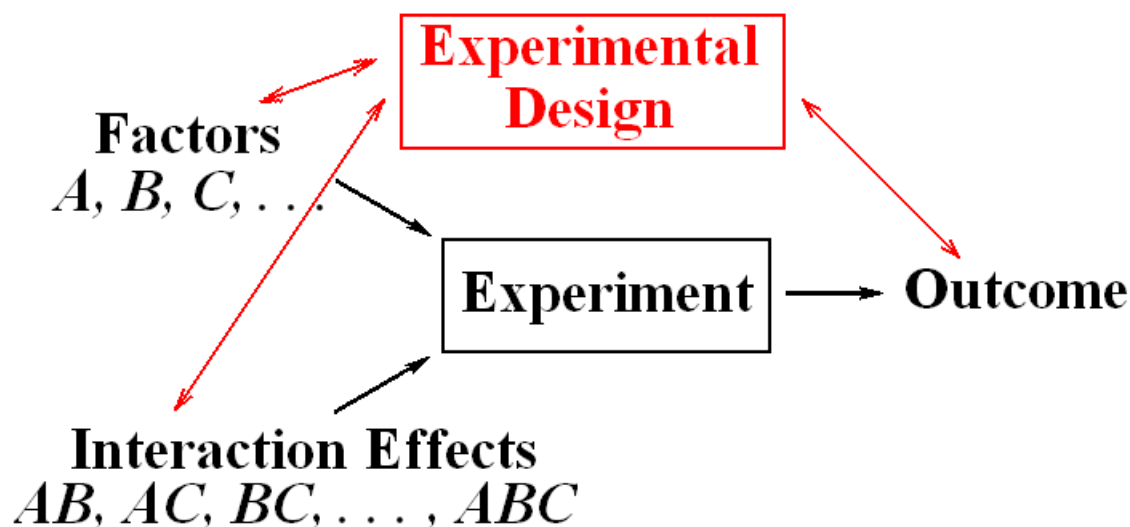


Fig 1. Schematic of Experimental Design.

The schematic of experimental design is shown in Fig. 1 above. Experimental design is useful in interpreting the outcome of an experiment. It provides an efficient identification of key variables that affect the outcome of the experiment, including interaction effects that are often not immediately obvious among the factors under study. It makes possible an efficient establishment of causality relationship between the factors and their interaction effects and the outcome of the experiment. One could attempt to determine the effects of various factors on the outcome of the experiment by varying the factors one at a time with the misguided notion that the interpretation of the experimental outcome would be simpler, but such an approach would be inefficient and it could lead to erroneous conclusions because it overlooks possible interaction effects among the various factors that could be important. A case in point will be shown shortly.

The benefits of experimental design have been convincingly demonstrated in the following case studies using the simplest application of experimental design: (1) using a 2-level, 4-factor (2 x 4) experimental design, the *Tool Products Company*, Plymouth, was able to pinpoint the cause of

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defects in the manufacturing of die-cast aluminum housings for computer disk drives (specifically, metal temperature and final die pressure acting together) and thereby reduce the incidence of defects from 15% to essentially zero¹; (2) with the aid of a 2-level, 6-factor (2 x 6) experimental design, engineers at Eastman Kodak were able to retool an existing metal-forming machine with the net results of a higher production rate of camera film clip that it produced with a greater reliability and a 10-fold scrap reduction²; and (3) with the help of a 2-level, 3-factor (2 x 3) experimental design, the design engineers of SKF (a Swedish company) were able to optimize the design of the ball bearing it produced and thereby successfully met the competition from a Japanese rival³.

The last case study is presented in details below to illustrate generally the analysis of a 2-level, n-factor (2 x n) experimental design. A two-level factorial design experiment means that each of the factors under study is varied to take on two assigned values (denoted by the plus- and minus-values). A three-level experiment means three assigned values (denoted by plus-, mid- and minus-values) for each of the variables being studied. Performance experience suggested that three factors affected the effective lifetime of a ball bearing, specifically, (i) outer ring osculation (see the figure below), (ii) heat treatment of the inner ring (see Fig 2 below), and (iii) material of construction for the bearing cage.

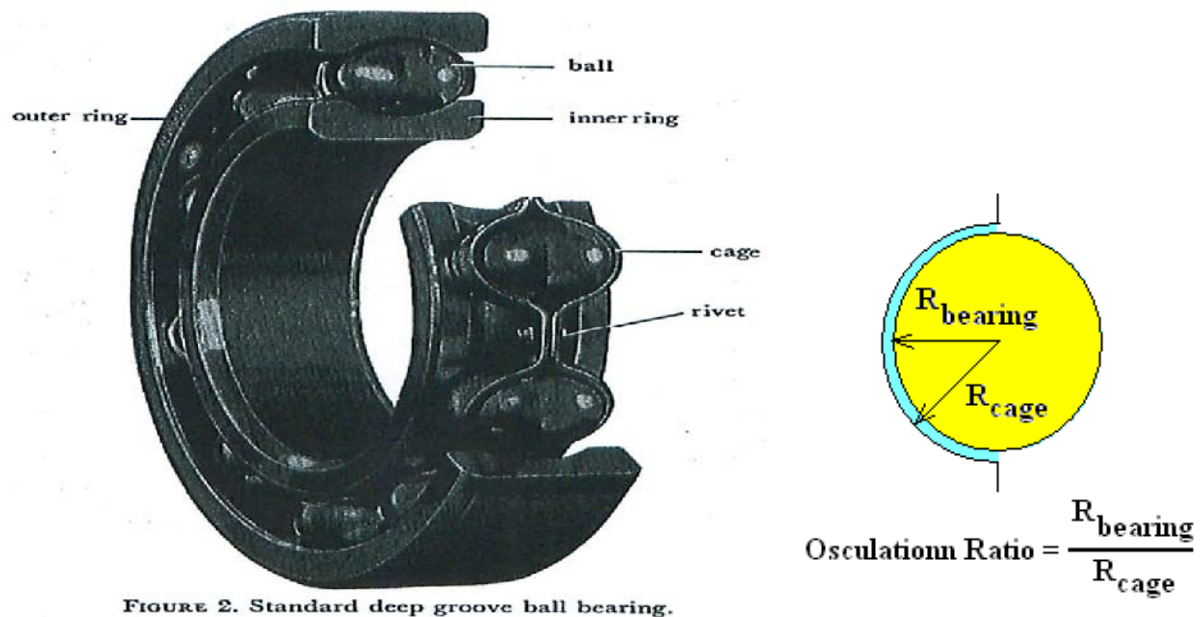


FIGURE 2. Standard deep groove ball bearing.

In the 2 x 3 factorial design study performed by SKF engineers, the osculation ratio was varied between a standard value and a modified value, the heat treatment of the inner ring was varied between standard treatment and an enhanced treatment, and the material of construction was varied between (cheaper) polymer and steel. A total of 2³ or 8 runs were performed to measure the effective lifetime of the ball bearing (presumably under accelerated test conditions) for the different parametric combinations, and the test results are presented in Table 1 on the next page.

The test results revealed two unexpected findings. First, unusually strong interaction effects were found among the test factors, especially between the outer ring osculation and the inner ring

heat treatment. Second, the material of construction for the bearing cage had only modest effect on the bearing life, suggesting the possibility of a major cost saving with the use of polymer material with only minor impact on the bearing's service life. These findings are supported by the following analysis.

Table 1. Effective lifetime of ball bearing under various test conditions.

Run #	Test Parameters			Measured Lifetime Y, h
	Osculation Ratio (A)	Inner Ring Heat Treatment (B)	Cage Material (C)	
1	Standard (A-)	Standard (B-)	Polymer (C-)	17
2	Modified (A+)	Standard (B-)	Polymer (C-)	25
3	Standard (A-)	Enhanced (B+)	Polymer (C-)	26
4	Modified (A+)	Enhanced (B+)	Polymer (C-)	85
5	Standard (A-)	Standard (B-)	Steel (C+)	19
6	Modified (A+)	Standard (B-)	Steel (C+)	21
7	Standard (A-)	Enhanced (B+)	Steel (C+)	16
8	Modified (A+)	Enhanced (B+)	Steel (C+)	128

$$\text{Overall Mean } \langle Y_{A,B,C} \rangle = \sum_{A=A-}^{A+} \sum_{B=B-}^{B+} \sum_{C=C-}^{C+} Y_{A,B,C} = [(17+25+26+85+19+21+16+128)/8] h = 42.125 h$$

$$\begin{aligned} \text{Effect of A} &\equiv \text{Mean } Y_{A+,B,C} - \text{Mean } Y_{A-,B,C} \\ &= [(25+85+21+128)/4 - (17+26+19+16)/4] h \\ &= 45.25 h \end{aligned}$$

$$\begin{aligned} \text{Effect of B} &\equiv \text{Mean } Y_{A,B+,C} - \text{Mean } Y_{A,B-,C} \\ &= [(26+85+16+128)/4 - (17+25+19+21)/4] h \\ &= 43.25 h \end{aligned}$$

$$\begin{aligned} \text{Effect of C} &\equiv \text{Mean } Y_{A,B,C+} - \text{Mean } Y_{A,B,C-} \\ &= [(19+21+16+128)/4 - (17+25+26+85)/4] h \\ &= 7.75 h \end{aligned}$$

Effect of AB Interaction

$$\begin{aligned} &= \text{Half difference between A effect when B is B+ and the A effect when B is B-} \\ &= \text{Half difference between B effect when A is A+ and the B effect when A is A-} \\ &= [(\text{Mean } Y_{A+B+C} - \text{Mean } Y_{A-B+C}) - (\text{Mean } Y_{A+B-C} - \text{Mean } Y_{A-B-C})]/2 \\ &= \{[(Y_{A+B+C+} + Y_{A+B+C-}) - (Y_{A-B+C+} + Y_{A-B+C-})] - [(Y_{A+B-C+} + Y_{A+B-C-}) - (Y_{A-B-C+} + Y_{A-B-C-})]\}/4 \\ &= \{[(128 + 85) - (16+26)] - [(21+25) - (19+17)]\}h/4 \\ &= \{171 - 10\}h/4 \\ &= 40.25 h \end{aligned}$$

Effect of AC Interaction

$$\begin{aligned} &= \text{Half difference between A effect when C is C+ and the A effect when C is C-} \\ &= \text{Half difference between C effect when A is A+ and the C effect when A is A-} \\ &= [(\text{Mean } Y_{A+BC+} - \text{Mean } Y_{A-BC+}) - (\text{Mean } Y_{A+BC-} - \text{Mean } Y_{A-BC-})]/2 \end{aligned}$$

$$\begin{aligned}
&= \{[(Y_{A+B+C+} + Y_{A+B-C+}) - (Y_{A-B+C+} + Y_{A-B-C+})] - [(Y_{A+B+C-} + Y_{A+B-C-}) - (Y_{A-B+C-} + Y_{A-B-C-})]\}/4 \\
&= \{[(128+21) - (16+19)] - [(85+25) - (26+17)]\}h/4 \\
&= \{114 - 67\}h/4 \\
&= 11.75 h
\end{aligned}$$

Effect of *BC* Interaction

$$\begin{aligned}
&= \text{Half difference between } B \text{ effect when } C \text{ is } C+ \text{ and the } B \text{ effect when } C \text{ is } C- \\
&= \text{Half difference between } C \text{ effect when } B \text{ is } B+ \text{ and the } C \text{ effect when } B \text{ is } B- \\
&= [(\text{Mean } Y_{AB+C+} - \text{Mean } Y_{AB-C+}) - (\text{Mean } Y_{AB+C-} - \text{Mean } Y_{AB-C-})]/2 \\
&= \{[(Y_{A+B+C+} + Y_{A-B+C+}) - (Y_{A+B-C+} + Y_{A-B-C+})] - [(Y_{A+B+C-} + Y_{A-B+C-}) - (Y_{A+B-C-} + Y_{A-B-C-})]\}/4 \\
&= \{[(128+16) - (21+19)] - [(85+26) - (25+17)]\}h/4 \\
&= \{104 - 69\}h/4 \\
&= 8.75 h
\end{aligned}$$

Effect of *ABC* Interaction

$$\begin{aligned}
&= \text{Half difference between the } AB \text{ interactions when } C \text{ is } C+ \text{ and } C \text{ is } C- \\
&= \{[(Y_{A+B+C+} - Y_{A-B+C+}) - (Y_{A+B-C+} - Y_{A-B-C+})]/2 - [(Y_{A+B+C-} - Y_{A-B+C-}) - (Y_{A+B-C-} - Y_{A-B-C-})]/2\}/2 \\
&= [Y_{A+B+C+} + Y_{A-B+C+} + Y_{A+B-C+} + Y_{A-B-C+}]/4 - [Y_{A+B+C-} + Y_{A-B+C-} + Y_{A+B-C-} + Y_{A-B-C-}]/4 \\
&= \{[128 + 19 + 26 + 25] - [16 + 21 + 85 + 17]\}h/4 \\
&= 14.75 h
\end{aligned}$$

It is seen that factors *A* (outer ring osculation) and *B* (inner ring heat treatment) and their interaction (*AB*) have much larger effects than factor *C* (material of cage construction).

The 2-level factorial design provides a model equation that not only summarizes the test results but also predict model results that could be tested for different parameter settings within the ranges tested. For the above example, the model equation predicting the bearing lifetime of is

$$\text{given by } Y_{\text{model}} = \beta_0 + \beta_1 A + \beta_2 B + \beta_3 C + \beta_4 A*B + \beta_5 A*C + \beta_6 B*C + \beta_7 A*B*C \quad (1)$$

where β_0 = overall mean lifetime = 42.125 h,

β_1 = *A* Effect/ $(A^+ - A^-)$ = 45.25 h/2 = 22.625 h, if $A^+ = +1$ and $A^- = -1$,

β_2 = *B* Effect/ $(B^+ - B^-)$ = 43.25 h/2 = 21.625 h, if $B^+ = +1$ and $B^- = -1$,

β_3 = *C* Effect/ $(C^+ - C^-)$ = 7.75 h/2 = 3.875 h, if $C^+ = +1$ and $C^- = -1$,

β_4 = *AB* Effect/2 = 40.25 h/2 = 20.125 h,

β_5 = *AC* Effect/2 = 11.75 h/2 = 5.875 h,

β_6 = *BC* Effect/2 = 8.75 h/2 = 4.375 h, and

β_7 = *ABC* Effect/2 = 14.75 h/2 = 7.375 h

It follows that Eqn (1) =>

$$Y_{\text{model}} = [42.125 + 22.625A + 21.625B + 3.875C + 20.125AB + 5.875AC + 4.375BC + 7.375ABC] h \quad (1')$$

The following table may be constructed using Eqn (1').

Run #	Test Parameters			Measured Lifetime Y, h	Model Lifetime Y_{models} h
	A	B	C		
1	A- = -1	B- = -1	C- = -1	17	17
2	A+ = 1	B- = -1	C- = -1	25	25
3	A- = -1	B+ = 1	C- = -1	26	26
4	A+ = 1	B+ = 1	C- = -1	85	85
5	A- = -1	B- = -1	C+ = 1	19	19
6	A+ = 1	B- = -1	C+ = 1	21	21
7	A- = -1	B+ = 1	C+ = 1	16	16
8	A+ = 1	B+ = 1	C+ = 1	128	128
9	-0.5	0	-1		29.9
10	0.75	-0.5	1		44.06

Red entries are model predictions that could be tested.

References

1. "Diecaster achieves zero-defect parts," *Quality in Manufacturing* (March/April, 1994).
2. Peter Runke, "Design of Experiments Software Saves Kodak Thousands," *MetalForming OnLine*, \DesignExpt\Design of Experiments Software Saves Kodak Thousands.mht.
3. C. Hellstrand, A. D. Oosterhoom, D. J. Sherwin and M. Gerson, "The Necessity of Modern Quality Improvement and Some Experience with its Implementation in the Manufacture of Rolling Bearings," *Phil Trans, Royal Soc. London A* 327, 529-537 (1989).